OFFICIAL RESEARCH DOSSIER | PROJECT REBOOT 2.0 | AI & IOT PUBLIC HEALTH ARCHITECTURE

AarogyaJal: AI-Powered Smart Water Health Surveillance & Early Warning System

A Proactive, Multi-Sensor Epidemiological Platform for Rural & Tribal Water Security

PRINCIPAL INVESTIGATOR	DOMAIN FOCUS	DEPLOYMENT TARGET	HARDWARE ARCHITECTURE
Manan Ramani	IoT, Edge AI & Syndromic Surveillance	Northeastern Region (NER) & Rural India	ESP32 / Multi-Sensor Edge Node

38+ Million	~200,000	< ₹5,500	24/7 + SMS
Annual Water-Borne Cases (India)	Preventable Annual Deaths	Autonomous Sensor Node Cost	Offline & Intermittent Resilient

1. Executive Summary & Problem Landscape

Water-borne diseases—predominantly diarrhea, cholera, typhoid, and hepatitis A—present a severe, recurring mortality burden across rural and tribal belts in India, most acutely throughout the Northeastern Region (NER). Fragile riverine topography, unsealed community borewells, open drainage runoff, and severe monsoon inundations regularly trigger microbial and chemical infiltration into primary drinking water sources.

Currently, public health management functions in a **reactive posture**: disease outbreaks are identified only after rural Primary Health Centres (PHCs) experience an influx of acutely ill patients. Due to mountainous terrain and isolated transit corridors, medical and chlorination responses arrive days or weeks late, allowing localized water contamination to multiply into community-wide epidemics.

The Critical Surveillance Bottleneck
Conventional microbiological lab testing requires physical sample carriage and 24–72 hours of culturing. AarogyaJal replaces this lagging model with an **integrated early warning paradigm** combining continuous physical-chemical telemetry (pH, Turbidity, TDS, Temp, Level, DO, Volatile Gases), rapid weekly field test strips (E. coli / Coliforms), and community syndromic reporting via ASHA health workers.

2. Multi-Vector Disease, Pathogen & Sensor Indicator Matrix

AarogyaJal establishes direct correlations between epidemic pathogens and measurable physical/chemical proxies that can be reliably tracked using low-cost sensor nodes:

DISEASE TARGET	PRIMARY PATHOGEN & VECTOR	KEY WATER CONTAMINATION PROXIES	SENSOR & DETECTION MODALITY
Diarrhea	*Escherichia coli*, Rotavirus, Cryptosporidium	Turbidity surges, pH shifts, fecal sediment intrusion	Optical Turbidity Sensor, Analog pH Module, *E. coli* Test Strips
Cholera	*Vibrio cholerae* (fecal-oral pathway)	Coliform proliferation, organic decay, dissolved oxygen depletion	Dissolved Oxygen (DO) Sensor, Turbidity Sensor, Rapid Coliform Strips
Typhoid	*Salmonella typhi* (sewage intrusion)	Conductivity surges, elevated Total Dissolved Solids, fecal traces	TDS / Electrical Conductivity (EC) Probe, Rapid Field Strips
Hepatitis A	Hepatitis A virus (HAV)	Fecal contamination proxies, sewage backflow, abnormal pH	pH Sensor, Turbidity Module, Coliform Biosensor Strips
Septic / Stagnant	Anaerobic bacterial bloom, rotting matter	Severe DO drop, thermal spikes, toxic gas liberation (H_2S, NH_3)	DO Sensor, Temp (DS18B20), MQ-136 (H_2S), MQ-137 (Ammonia)

3. Multi-Tier Sensing Pipeline & Sensor Modality Trade-Off Analysis

Deploying continuous water surveillance across rugged, low-resource terrain requires an explicit engineering compromise between measurement accuracy, capital cost, and maintenance tolerance.

1
Continuous IoT
24/7 continuous telemetry of pH, Level, TDS, Temp, Turbidity, DO, and headspace gases.

2
ASHA Telemetry
ASHA health workers log weekly coliform test strips and cluster symptoms via mobile app/SMS.

3
AI Outbreak Engine
Ensemble ML cross-correlates parameter drift with syndromic logs and monsoon flooding data.

4
Rapid Response
Automated SMS advisories dispatch to PHCs, district officers, and village water committees.

3.1 Sensor Modality Trade-Offs: Servicing Tolerance vs. Budget

The central architectural dilemma in environmental monitoring is: *How much manual servicing can this deployment tolerate vs. budget?* In rural Northeastern India, where nodes are hours away from district centers, automated unattended reliability is paramount.

ISE (Ion-Selective) — Drift-Prone

Pros: Cheap and simple for specific ions (Nitrate/Fluoride).
Cons: Severe calibration drift; highly sensitive to ionic interference. Requires recurring calibration against known chemical standards, making unattended rural operation fragile.

Wet-Chemical Analyzers — Reagent-Heavy

Pros: Gold-standard lab accuracy in the field.
Cons: Consumes liquid reagents. Requires a trained field technician for periodic refills and pump maintenance. Fails the "install-and-forget" requirement in remote tribal terrain.

Optical / UV Absorption — High CapEx

Pros: Drift-free, highly accurate, low ongoing maintenance.
Cons: Prohibitively expensive per unit (₹40,000–₹1,50,000+ per probe). Cannot scale across thousands of rural panchayats with limited public health budgets.

AarogyaJal's Winning Design Decision: Solid-State Core + Surrogate AI Modeling

Instead of fragile wet-chemical reagents or cost-prohibitive UV probes, AarogyaJal deploys robust, solid-state sensors (**pH, Water Level, TDS, Temperature, Turbidity, DO**) paired with periodic weekly manual paper test strips. This eliminates recurring technician reagent costs while retaining maximum field durability.

3.2 Real-Time Surrogates: BOD & Algal Bloom (Chlorophyll-a)

- **Biochemical Oxygen Demand (BOD):** BOD cannot be sensed directly in real-time because standard BOD_5 requires a 5-day biological incubation period. AarogyaJal circumvents this by computing an *Inferred Organic Load Surrogate*: combining real-time **Dissolved Oxygen (DO) depletion rates**, **turbidity spikes**, and **water temperature** into an ML-derived proxy for sudden organic/septic contamination.
- **Chlorophyll-a (Algal Bloom Proxy):** Measured via fluorescence sensors. Evaluated as an optional Phase-2 add-on for open surface reservoirs and community harvesting ponds, but excluded from base drinking water nodes to preserve sub-₹5,500 unit economics.
- **The Fundamental Four Continuous Baseline: pH** (chemical stability), **Water Level** (tank volume/sudden drainage), **TDS/Conductivity** (salinity/dissolved ionic intrusion), and **Temperature** (thermal correction & bacterial kinetic modeling).

4. IoT Hardware Bill of Materials (BOM) & Unit Economics

The AarogyaJal hardware stack is engineered around widely accessible, commercially available industrial and hobby-grade components sourced directly from Indian supply chains, achieving unprecedented unit economics:

COMPONENT	PRICE RANGE (INR)	SUPPLIER REFERENCES	TECHNICAL SPECIFICATION & FUNCTION
pH Sensor Kit	₹800 – ₹1,500	QBM India / Graylogix	Analog glass probe + signal conditioning board (0–14 pH, ±0.1 accuracy).
Turbidity Sensor Module	₹500 – ₹600	Robu.in / HatchnHack	Optical infrared phototransistor measuring light scatter (0–3000 NTU).
TDS / Conductivity Sensor	₹250 – ₹450	Robu / QuartzComponents	Dual-pin AC excitation probe preventing electrode polarization (0–1000 ppm).
Water Temperature Sensor	₹120 – ₹220	Quartz / Robu	DS18B20 digital stainless steel waterproof probe (-55°C to +125°C).
Water Level Sensor	₹80 – ₹200	HatchnHack / Quartz	JSN-SR04T waterproof ultrasonic probe for reservoir tank volume monitoring.
Dissolved Oxygen (DO) Probe	₹1,000 – ₹2,500	Robu / Importers	Galvanic/polarographic DO probe detecting organic decay and stagnation.
Gas & Odour Sensor Suite	₹100 – ₹300	Quartz / Robu	MQ-136 (H_2S), MQ-137 (Ammonia) to detect anaerobic organic decomposition.
ESP32 / NodeMCU MCU	₹220 – ₹350	QuartzComponents / Robu	Dual-core 240MHz, Wi-Fi 802.11 b/g/n + BLE, deep sleep power cycling (<15µA).
Power Subsystem (Battery)	₹400 – ₹1,500	Local / Li-ion Cells	3.7V 18650 Li-ion / Li-Po rechargeable pack with BMS protection circuit.
Solar Harvesting Panel	₹600 – ₹2,500	Solar Retailers / Robu	5W–10W monocrystalline solar panel + TP4056 / MPPT step-down solar charger.
Weatherproof Enclosure	₹350 – ₹700	IP66 Junction Boxes	UV-resistant ABS enclosure with IP68 cable glands and mounting bracket.
TOTAL COMPLETE NODE	**₹4,420 – ₹9,320**	**~$53 – $110 USD**	**96%+ cost reduction vs. commercial stations (₹1.5 Lakh – ₹5 Lakh)**

Mass Deployability & Fiscal Sustainability

At an average node cost of **₹5,500 (~$65 USD)**, an entire district water surveillance network spanning 50 critical points of interest (handpumps, storage tanks, and riverbanks) can be outfitted for under **₹2.75 Lakh**—less than the acquisition cost of a single municipal-grade SCADA monitoring station.

4.1 Edge Energy Budget, Harvesting & Anti-Fouling Architecture

- **Low-Power Duty Cycle:** ESP32 firmware executes a 55-second deep sleep cycle (<15 µA quiescent draw) followed by a 5-second burst sensing, analog-to-digital averaging, and MQTT transmission phase. Average operational consumption is constrained to ~20 mA at 3.3V.
- **Monsoon Solar Power Budget:** A compact 5W–10W monocrystalline solar panel produces 18–25 Wh daily under normal diffuse sunlight. The edge node consumes approximately 0.48 Wh per 24-hour operating period, providing a ~40× safety multiplier to sustain autonomous operation through 14 consecutive days of heavy overcast monsoonal rains.
- **Anti-Fouling & Electrode Longevity:** Alternating current (AC) excitation pulses across TDS/conductivity probes eliminate galvanic polarization, prevent electrode oxidation, and inhibit biological biofilm fouling in stagnant well environments.

5. Target Pilot Deployments: Northeastern Region (NER)

Due to extreme rainfall, seasonal floods, and fragmented healthcare networks, five high-risk pilot sites in the NER have been earmarked to validate sensor performance and predictive outbreak modeling under harsh operating conditions:

1. Majuli, Assam — **River Island Aquifer**
Installation Point: Exit manifold of community handpump near Brahmaputra riverbank.
Risk Vector: Monsoonal river swells cause subsurface silt infiltration and cross-contamination of shallow drinking aquifers.

2. Churachandpur, Manipur — **Centralized Reservoir**
Installation Point: Village-level gravity-fed overhead storage reservoir.
Risk Vector: Centralized distribution point; a single pathogen infiltration instantly contaminates the entire settlement.

3. Tura, Meghalaya — **Traditional Open Well**
Installation Point: Community dug well located in Rongram Block.
Risk Vector: Heavy precipitation causes rapid agricultural and surface runoff into the well, spiking turbidity and fecal coliforms.

4. Aizawl, Mizoram — **Drainage Discharge**
Installation Point: Drain exit point discharging into a community harvesting pond.
Risk Vector: Untreated hillside sewage overflow mixes directly with primary community bathing and utility water.

5. Pasighat, Arunachal Pradesh — **Borewell Riverbank Infiltration**
Installation Point: Community borewell discharge outlet adjacent to the Siang riverbank.
Risk Vector: Frequent flash floods cause backflow and seepage of river silt and untreated surface waste into borewell intake pipes.

6. Comprehensive Industry & Competitive Benchmark

AarogyaJal combines the advantages of continuous IoT monitoring, microbiological field testing, and AI-driven syndromic fusion while maintaining accessibility for rural budgets:

COMPANY / SOLUTION	REAL-TIME IOT	PATHOGEN TESTING	AI / PREDICTIVE	DASHBOARD & ALERTS	RURAL / LOW-COST	COMPARATIVE TAKEAWAY
KETOS (US)	✓ Autonomous	• Auto-analyzer	✓ Predictive	✓ Cloud	✗ Industrial only	High-end automated analyzer; excessive cost for rural deployment.
Aquagenx	✗ Manual kits	✓ Coliform/E. coli	✗ None	• Manual logs	✓ Field-focused	Gold standard microbiological kits, but no continuous telemetry.
OmniVis	• Smartphone dock	✓ Cholera bio-test	✗ Research phase	• Mobile app	✓ Clinical scope	Rapid handheld cholera test, but requires manual operator sampling.
IntuThings (India)	✓ IoT modules	✗ No pathogens	• Telemetry stats	✓ Web portal	• Municipal focus	Indian IoT integrator focused on urban piped water networks.
CLUIX (India)	✓ Portable reader	• Optical proxies	✗ Limited	✓ App + Device	• Semi-portable	Portable digital water analyzer measuring 12+ physical parameters.
Hach / Xylem	✓ Industrial probes	• Lab methods	• SCADA analytics	✓ Enterprise UI	✗ High CapEx	Global benchmarks costing ₹15 Lakh – ₹50 Lakh per station.
s::can / Kando	✓ Spectrometry / IoT	• Optical proxies	✓ Event detection	✓ Real-time	✗ Utility only	Advanced optical spectrometry probes for municipal wastewater grids.
AquaSense (India)	✓ Surface / Ground	✗ No direct tests	• Anomaly alerts	✓ Web dashboard	• Commercial	Indian sensor solutions for environmental groundwater tracking.
Aqvify	✓ Level / TDS	✗ None	✗ Trend graphs	✓ Mobile App	✓ Well-level focus	Excellent telemetry for well depth, but lacks public health analytics.
AarogyaJal (Reboot)	**✓ Continuous Array**	**✓ Integrated Strips**	**✓ XGBoost / RF**	**✓ SMS + GIS Web**	**✓ Offline / <₹5.5k**	**Only hybrid model uniting low-cost IoT, ASHA syndromic logs, and predictive outbreak AI for rural geographies.**

7. Technical Architecture & Technology Stack

The AarogyaJal Reboot 2.0 system leverages a decoupled, offline-resilient microservice pattern built for high fault-tolerance across low-bandwidth environments:

Unified Technology Ecosystem

ESP32 / NodeMCU | MQTT (Mosquitto / EMQX) | FastAPI (Python) | Node.js / Express | PostgreSQL + PostGIS | TimescaleDB | MongoDB | Redis Caching | Next.js & React | Tailwind CSS | Flutter (Mobile App) | Scikit-learn | XGBoost | Gemini LLM (Advisory) | SMS Gateway / Twilio

SYSTEM LAYER	CORE TECHNOLOGIES	FUNCTIONAL RESPONSIBILITY
Hardware & Edge	ESP32 / NodeMCU, FreeRTOS, Arduino C++, Deep Sleep	Analog sampling, digital filtering, power management, burst MQTT telemetry.
Ingestion & Queuing	MQTT Broker (EMQX / Mosquitto), WebSockets, REST Gateways	High-throughput telemetry ingestion, buffering dropped packets during outages.
Core Backend API	FastAPI / NestJS, PostgreSQL + PostGIS, TimescaleDB, Redis	Time-series logging, spatial GIS queries, role-based access (PHC, ASHA, VWSC).
AI / ML Analytics	Python, XGBoost, Random Forest, Scikit-learn, Gemini API	Water Quality Index (WQI) scoring, anomaly detection, automated advisory alerts.
Web Dashboard & GIS	Next.js, React, Tailwind CSS, Leaflet / Mapbox GL	Geospatial contamination heatmaps, telemetry graphs, alert dispatch console.
Mobile Field Client	Flutter, SQLite (Offline-first), Background Sync	ASHA symptom logging, test strip camera capture, offline SMS fallback.

8. Outbreak Prediction Engine & Risk Index Formulation

The prediction engine computes a dynamic **Outbreak Risk Index (ORI)** by fusing physical telemetry deviations, microbiological validations, and localized syndromic health reports:

Algorithmic Formulation: Outbreak Risk Index (ORI)

$$ORI = w_1 \cdot \Delta WQI(t) + w_2 \cdot Biol(t) + w_3 \cdot \Sigma\, SymptomCluster(t, r) + w_4 \cdot SeasonRainFactor(t)$$

Where ΔWQI tracks rate of parameter deviation (pH, Turbidity, DO, TDS); Biol represents positive *E. coli* strip detection; SymptomCluster aggregates ASHA diarrheal and febrile complaint surges within radius *r*; and SeasonRainFactor scales risk during peak monsoon flooding. If ORI exceeds 0.75, automated emergency alerts are dispatched.

9. Reboot 2.0 Strategic Execution Roadmap

P1
Hardware & Edge
Refactor ESP32 firmware with burst MQTT, solar battery power management, and local calibration tables.

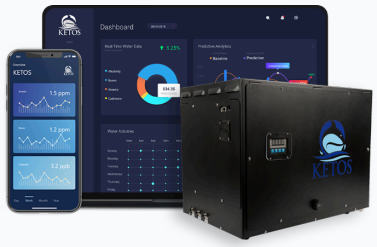
P2
Cloud Backend
Implement FastAPI backend with TimescaleDB time-series storage and ASHA symptom intake APIs.

P3
AI & Dashboard
Train XGBoost anomaly models, integrate WQI calculator, and build Next.js GIS surveillance console.

P4
Field Deployment
Pilot 5 nodes across Assam & Meghalaya, onboard ASHA workers, and benchmark prediction accuracy.

10. Visual Benchmark Gallery: Existing Market Hardware & Sensors

Visual inspection of commercial competitors across industrial kiosks, multiparameter probes, and handheld analyzers confirms the structural gap in rural accessibility:



KETOS SHIELD (US) — Autonomous Kiosk

Industrial-grade automated testing cabinet. Features automated robotic fluid handling, self-calibration, and multi-metal testing.

Limitation: Prohibitive CapEx; requires subscription SaaS & industrial housing.



YSI EXO1 Sonde (Xylem) — Multi-Probe Sonde

High-precision oceanographic & riverine sonde featuring 4 smart sensor ports, antifouling wiper, and titanium housing.

Limitation: Unit cost ₹8L–₹25L; designed for marine research, not village wells.



s::can spectro::lyser — UV-Vis Spectrometer

Submersible optical spectrometer measuring TOC, COD, BOD proxies, and nitrate absorption spectrum without reagents.

Limitation: Costs ₹15L–₹35L; high optical maintenance & compressed air cleaning.



OmniVis Cholera Device — Mobile Diagnostic

Purdue-developed smartphone hardware attachment using RT-LAMP isothermal amplification & particle diffusometry for cholera.

Limitation: Dedicated single-pathogen tool; requires manual sample extraction.



CLUIX C022 (India) — Handheld Analyzer

Indian IoT handheld digital analyzer measuring 12+ physical parameters with cloud sync and Jal Jeevan Mission integration.

Limitation: Operator-dependent grab testing; no continuous 24/7 telemetry.



Kando kLog V3.0 — Sewer Manhole IoT

Ruggedized underground controller for sewer trunk lines, detecting toxic municipal discharges and pathogen spikes via AI.

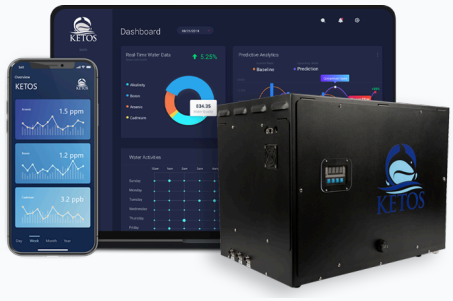
Limitation: Tailored strictly for municipal wastewater grids, not rural drinking aquifers.

The AarogyaJal Hardware Differentiation

Existing market hardware falls into two extremes: **hyper-expensive municipal SCADA instruments** (₹8L–₹35L like YSI and s::can) or **intermittent manual testing devices** (CLUIX, OmniVis). AarogyaJal creates a new category: an **autonomous 24/7 continuous multi-probe array at under ₹5,500** with integrated solar harvesting.

11. Visual Benchmark Gallery: Software Platforms & Field Applications

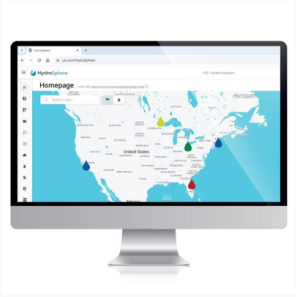
Examining competitor software architectures underscores the critical necessity of combining real-time environmental telemetry with bottom-up syndromic health reports:



KETOS Wave / Cloud Analytics — Enterprise SaaS

Enterprise cloud portal displaying real-time parameter tracking, automated compliance alerts, and multi-tenant telemetry dashboards.

Takeaway: High UI polish, but completely lacks grassroots syndromic (ASHA) health reporting.



YSI HydroSphere Platform — GIS River Network

Geospatial cloud platform mapping water sensors across river basins with automated thresholds, historical curves, and public portal view.

Takeaway: Excellent geospatial visualization; serves as benchmark for AarogyaJal's GIS map.



Aqvify Well Telemetry & Mobile App — Consumer IoT

Submersible pressure transducer paired with a mobile app displaying live water column height, 24h consumption graphs, and low-level alerts.

Takeaway: Validates the usability of mobile-first telemetry; lacks epidemiological disease predictive intelligence.



Aquagenx CBT Field Test Kit — Microbiological Field Protocol

WHO/UNICEF JMP validated compartment bag test quantifying *E. coli* and coliforms via MPN color change without electricity or laboratories.

Takeaway: Standardizes the exact biological spot-check protocol adopted for AarogyaJal's ASHA validation loop.

The AarogyaJal Synthesis & Architectural Monopoly

No single competitor bridges all three operational pillars: **(1) Sub-₹5,500 continuous physical IoT telemetry**, **(2) Rapid microbiological test-strip ground-truthing**, and **(3) Grassroots syndromic outbreak prediction via ASHA frontline workers**. AarogyaJal solves the rural water crisis by creating an interconnected epidemiological early-warning shield engineered precisely for India's most underserved regions.